



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECP2613 – SYSTEM ANALYSIS AND DESIGN

SEMESTER II 2023/2024

KADA MANAGEMENT INFORMATION SYSTEM

Analysis and Design

Phases:

DESIGN AND ANALYSIS

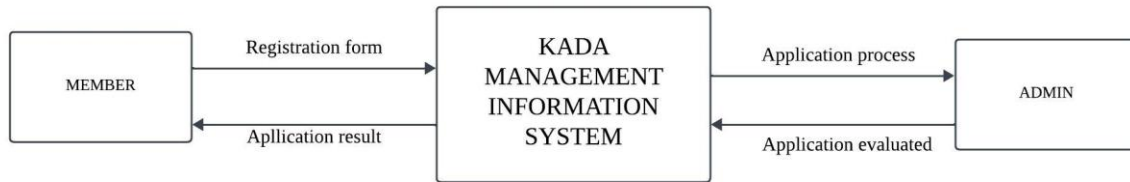
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Content

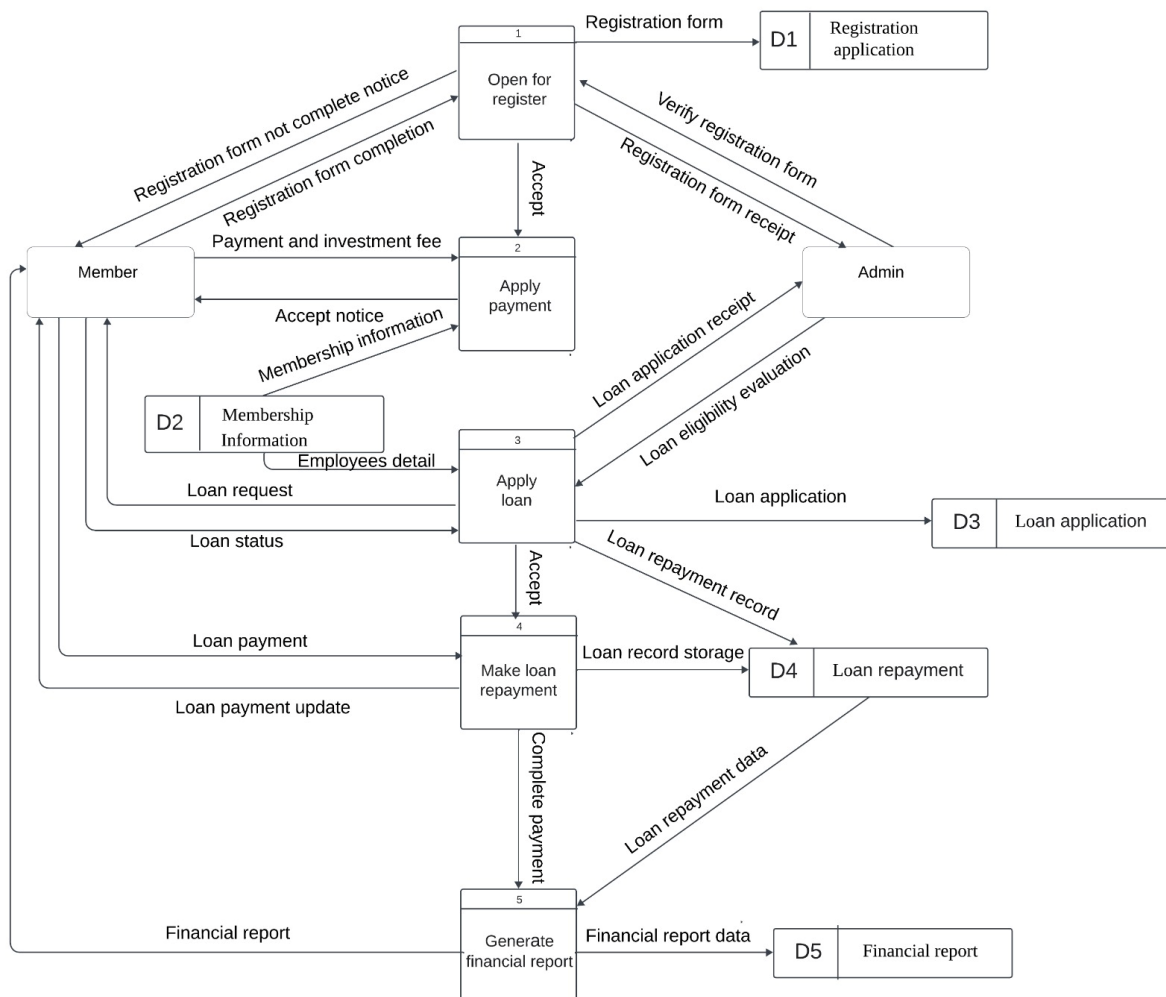
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1.0 Logical DFD (TO-BE) System

1.1 Context Diagram

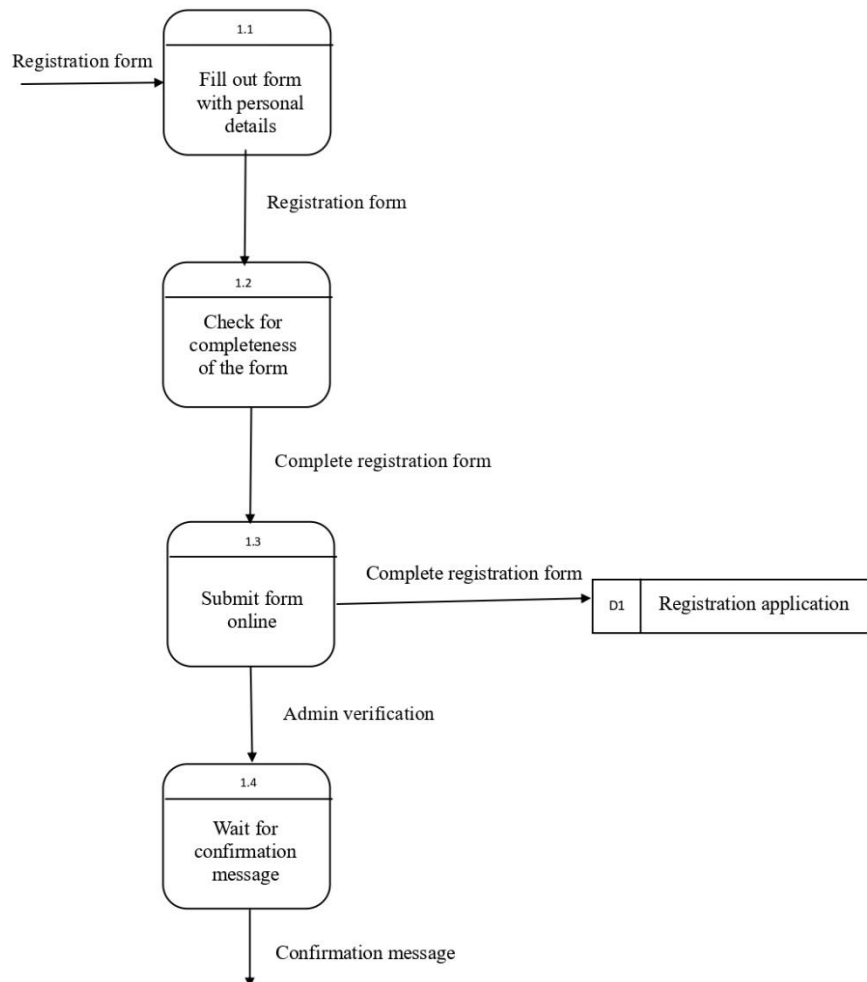


1.2 Diagram Zero

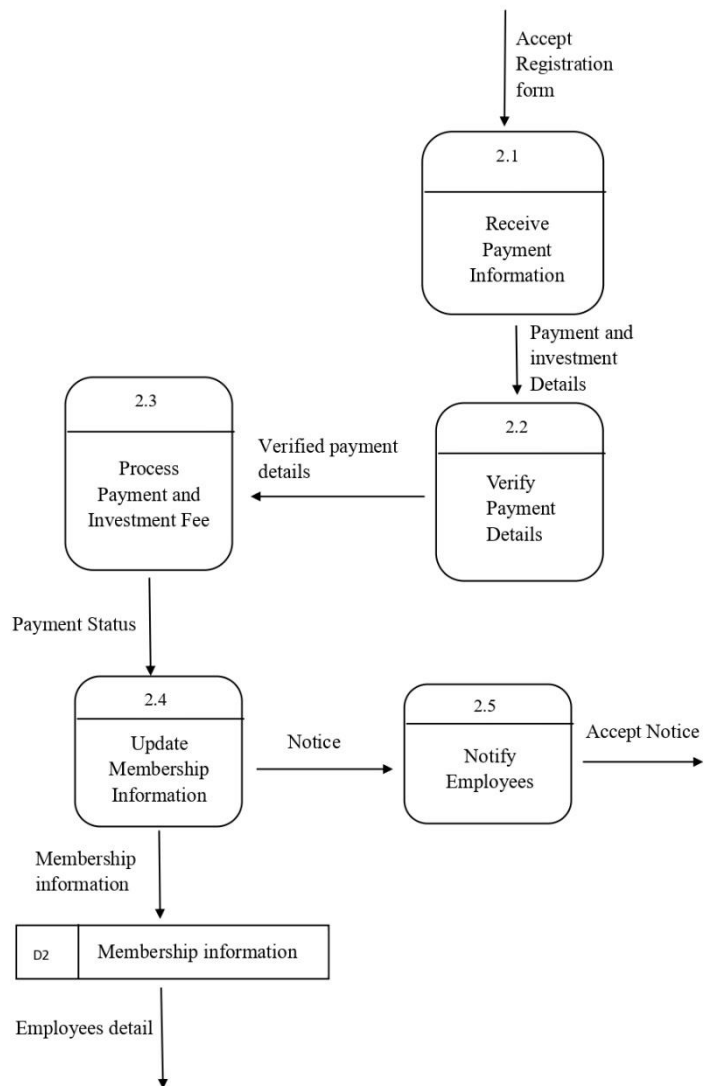


1.3 Child Diagram

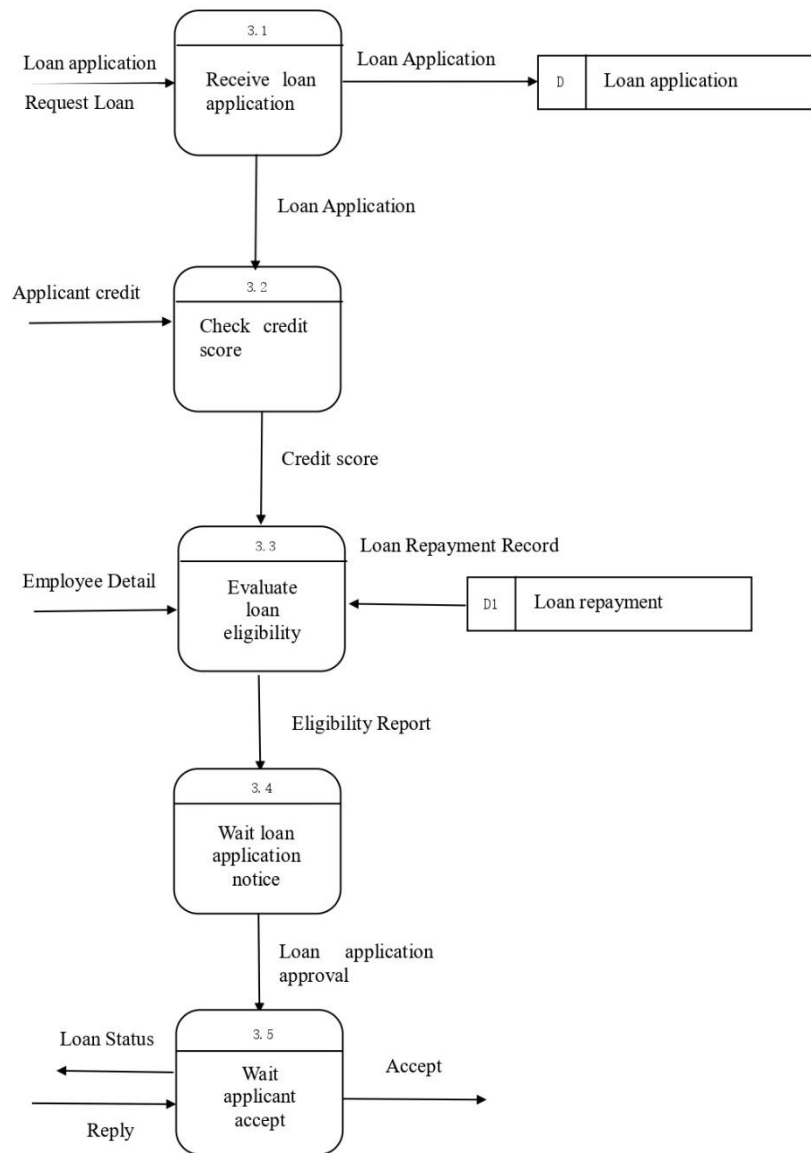
1.3.1 Open for register



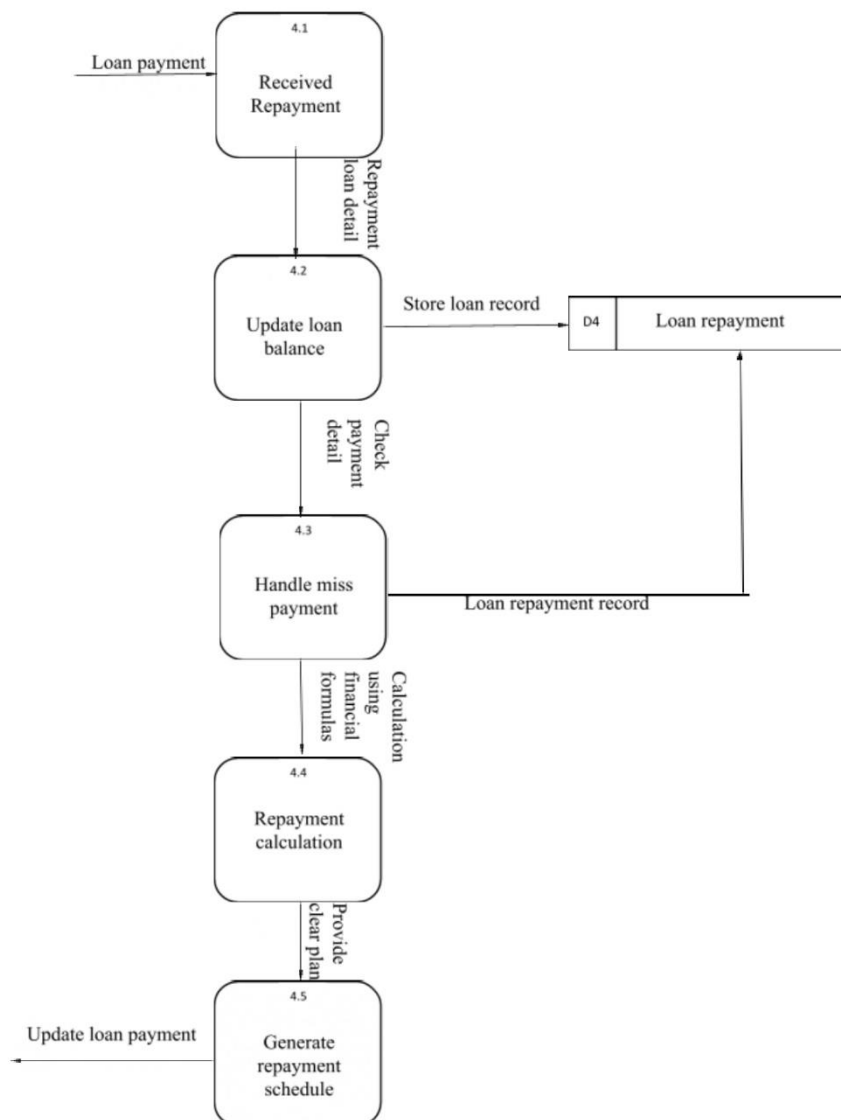
1.3.2 Apply payment



1.3.3 Apply loan



1.3.4 Make loan repayment



1.4 Process Identification (From Logical DFD(TO-BE) System)

Process 1: Open for register

DO

READ blank registration form from website

BEGIN IF

IF registration form is filled out with personal details

 Read filled registration form

THEN

 Check for completeness of the form

IF form is complete

 Submit the verified registration form to KADA through website

 Store filled form in database

ELSE

 Request additional information from the applicant

IF submission status is confirmed

 Update database with confirmation status

Perform notification of registration confirmation

Display confirmation message to applicant

ELSE

Display failure message to applicant

END IF

Process 2: Apply payment

DO READ the registration form

BEGIN IF

IF the registration form is received:

1. Extract payment details.
2. Verify payment.

THEN:

IF payment is valid:

1. Process payment.
2. **IF** payment is successful:
Confirm payment.
Update membership info.
Notify employees.

ELSE:

Notify about invalid payment.

END IF

Process 3: Apply loan

DO

Receive Loan Application

Validate Loan Application Data

BEGIN IF

IF Loan Application is Complete

 THEN Check Credit Score

ELSE

 Request Additional Information

ENDIF

IF Credit Score is Available

 THEN Proceed to Eligibility Evaluation

ELSE

 Send Credit Score Request Notification

ENDIF

IF Credit Score is Excellent

 Approve Loan

ELSE IF Credit Score is Good

 Approve Loan with Conditions

ELSE IF Credit Score is Fair

 Review Manually

ELSE IF Credit Score is Poor

 Reject Loan

ELSE

 Print Error

ENDIF

DO WHILE there are Pending Loan Applications

 Process Next Loan Application

 Validate Loan Application Data

IF Loan Application is Complete

 THEN Check Credit Score

ELSE

 Request Additional Information

ENDIF

IF Credit Score is Available

 THEN Proceed to Eligibility Evaluation

ELSE

Send Credit Score Request Notification
ENDIF

IF Credit Score is Excellent
 Approve Loan
ELSE IF Credit Score is Good
 Approve Loan with Conditions
ELSE IF Credit Score is Fair
 Review Manually
ELSE IF Credit Score is Poor
 Reject Loan
ELSE
 Print Error
ENDIF

ENDDO

Process 4: Make loan repayment

DO

READ loan payment

BEGIN IF

IF the complete loan payment is received

1. Store the repayment details to the loan repayment database.
2. Check the payment same with the due amount.

THEN:

Update the loan repayment record with the valid payment information.

Store the updated loan record in the database.

ELSE continue:

IF a missed payment is detected:

Perform the handling process for missed payments.

Perform the calculation of the new repayment schedule using financial formulas.

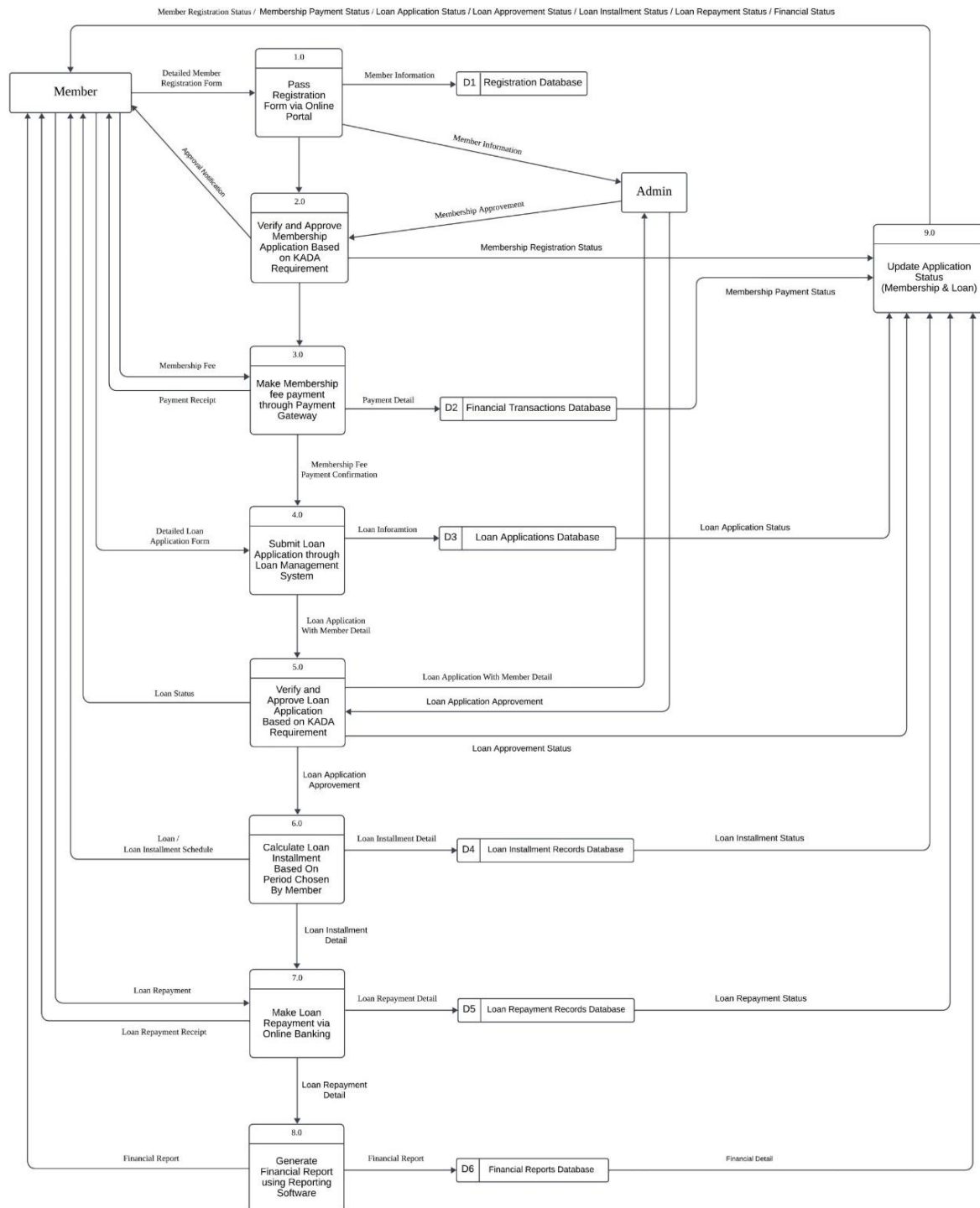
Notify the borrower about the missed payment

Display the new repayment schedule to the borrower

END IF

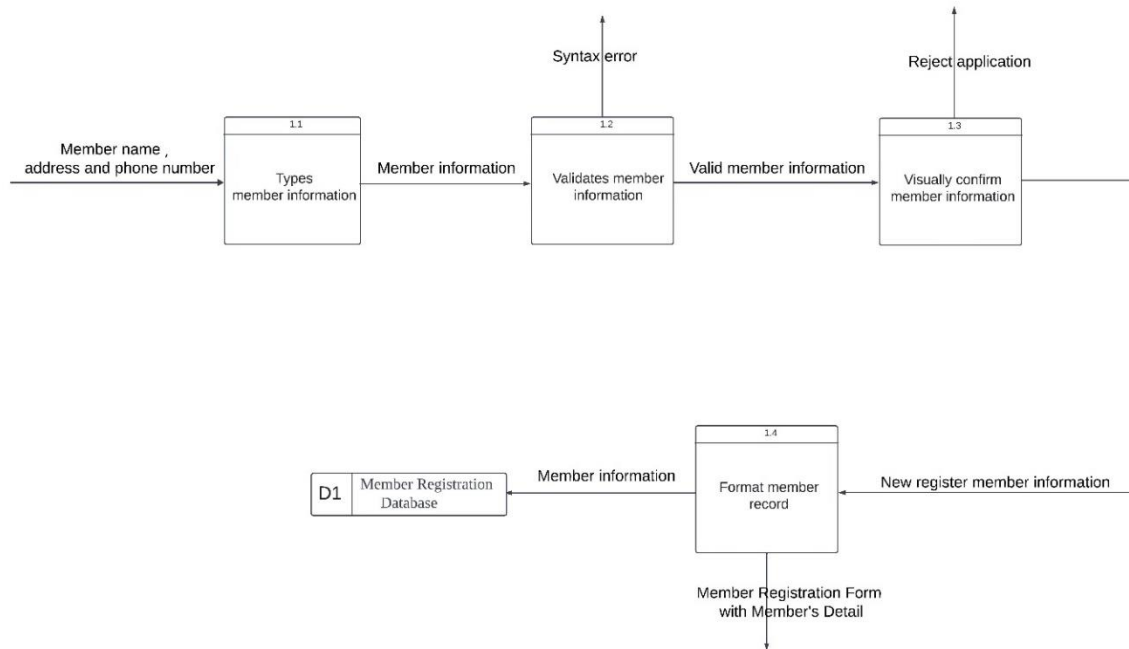
2.0 Physical DFD (TO-BE) System

2.1 Diagram Zero

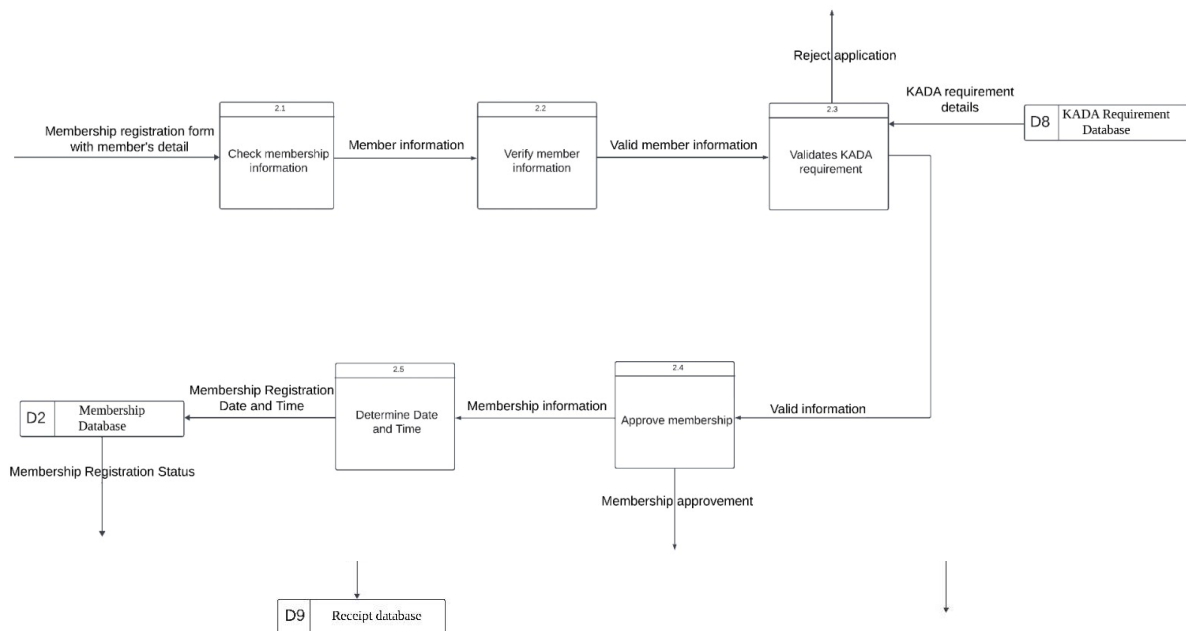


2.2 Child diagram

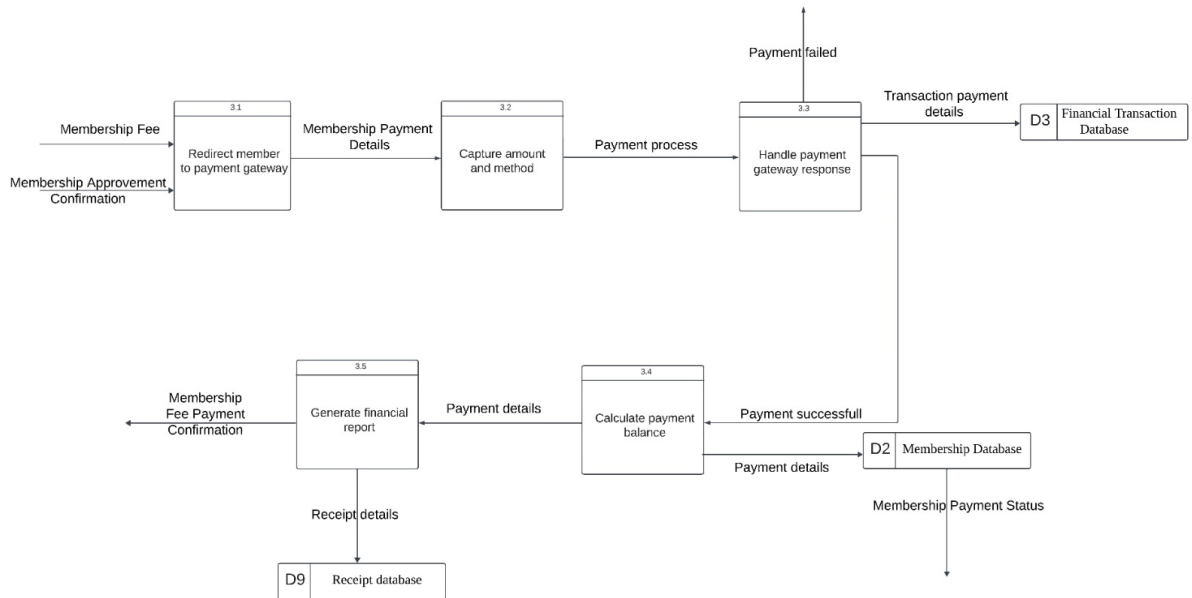
Child diagram 1



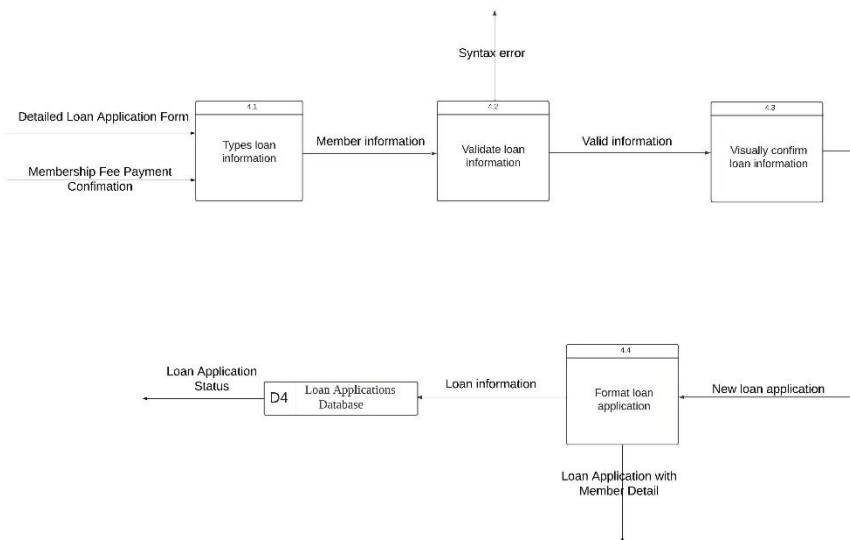
Child diagram 2



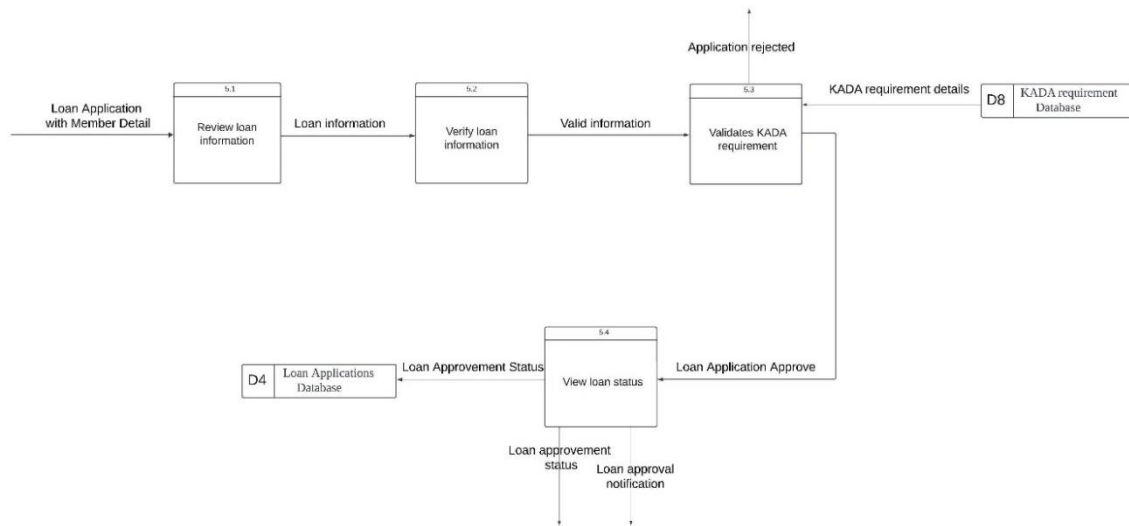
Child diagram 3



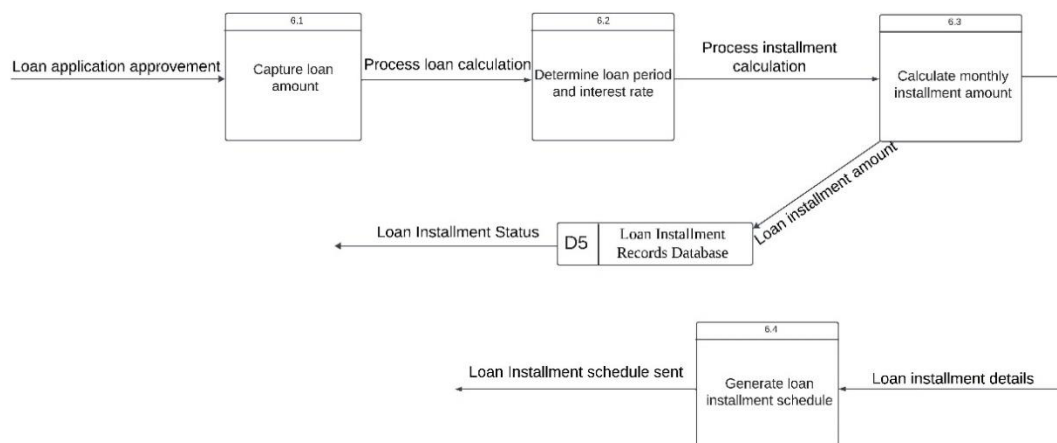
Child diagram 4



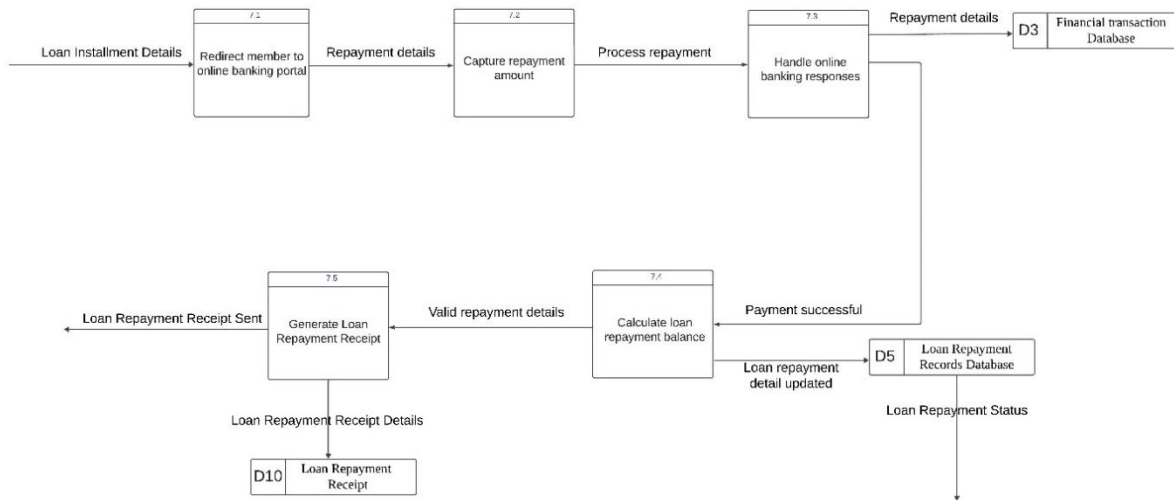
Child diagram 5



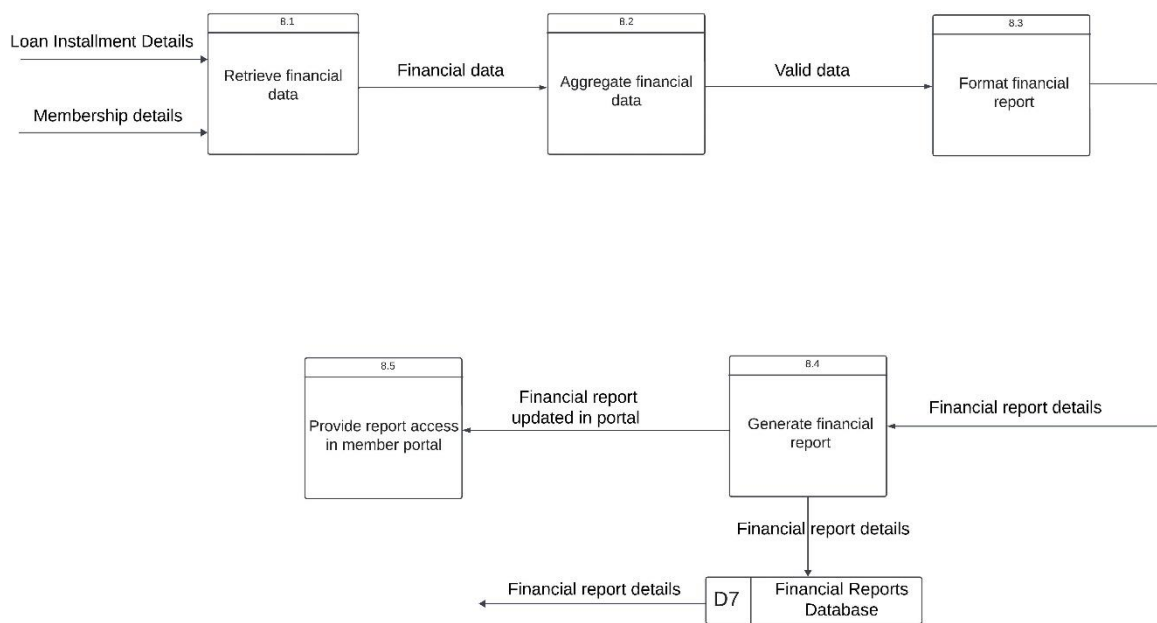
Child diagram 6



Child diagram 7



Child diagram 8

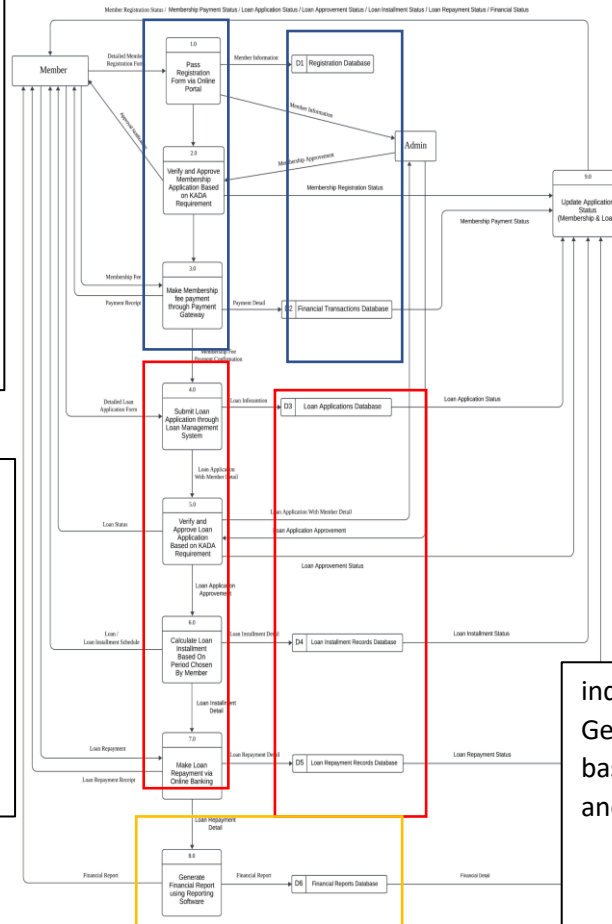


2.3 Partitioning

2.3.1 Diagram Zero

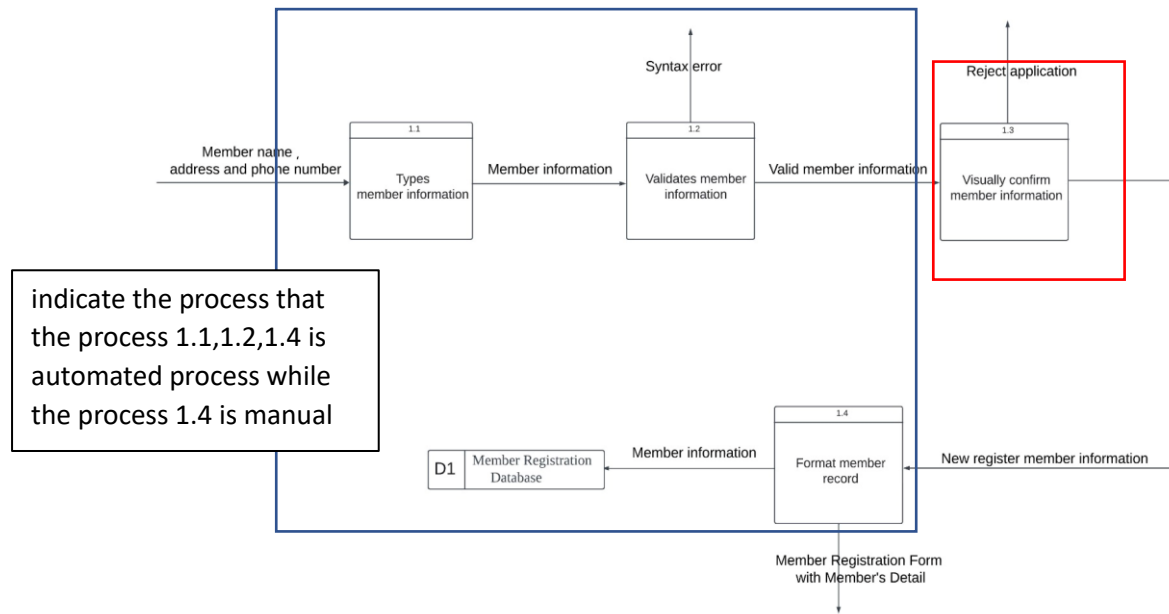
indicate the process that Handles member registration, verification, approval, and membership fee payment.

indicate the process that Manages loan applications, approvals, installment calculations, and repayment.

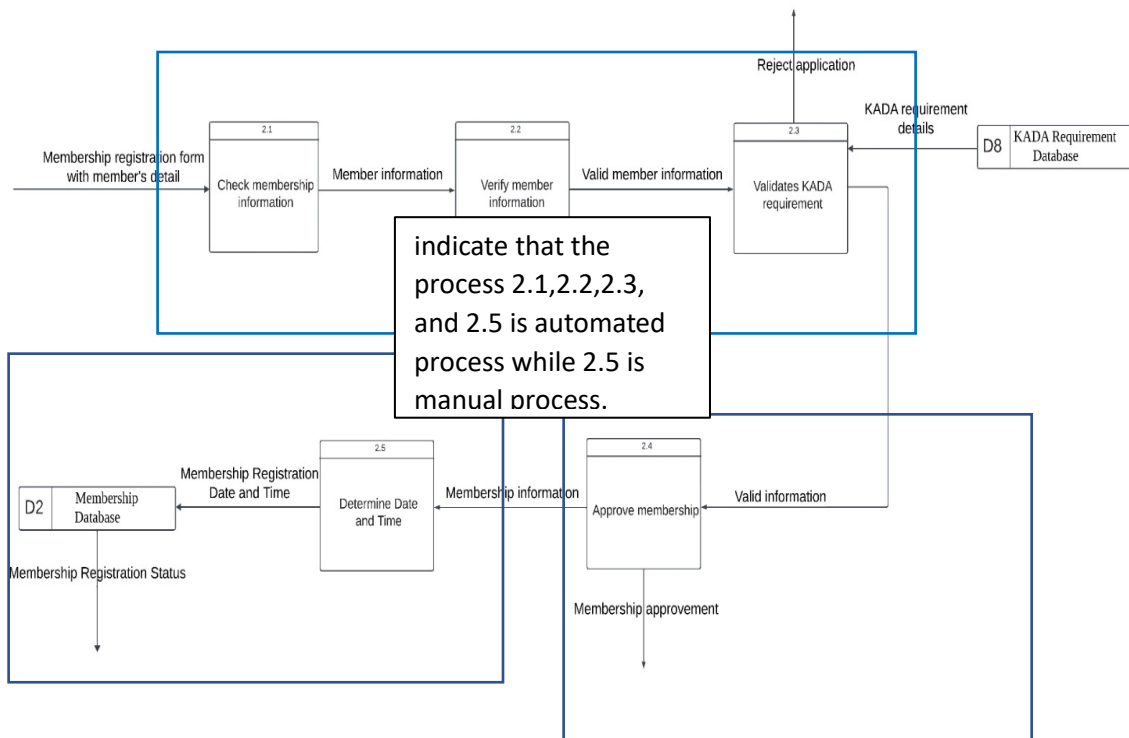


indicate the process that Generates financial reports based on transactions, loans, and repayments.

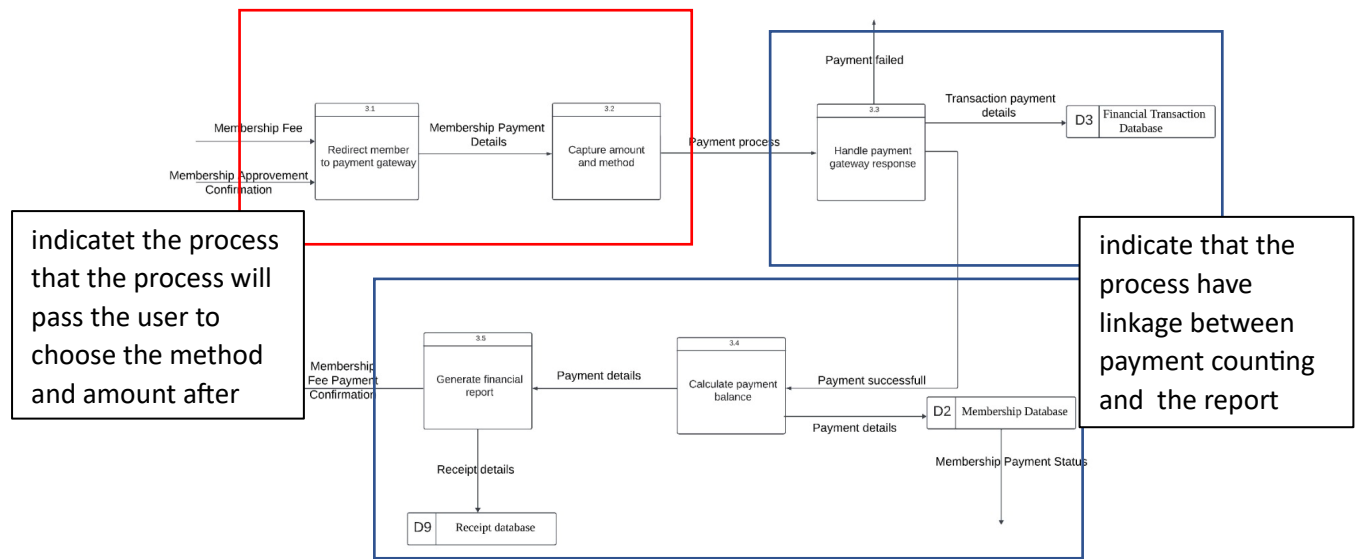
2.3.2 Process 1.0



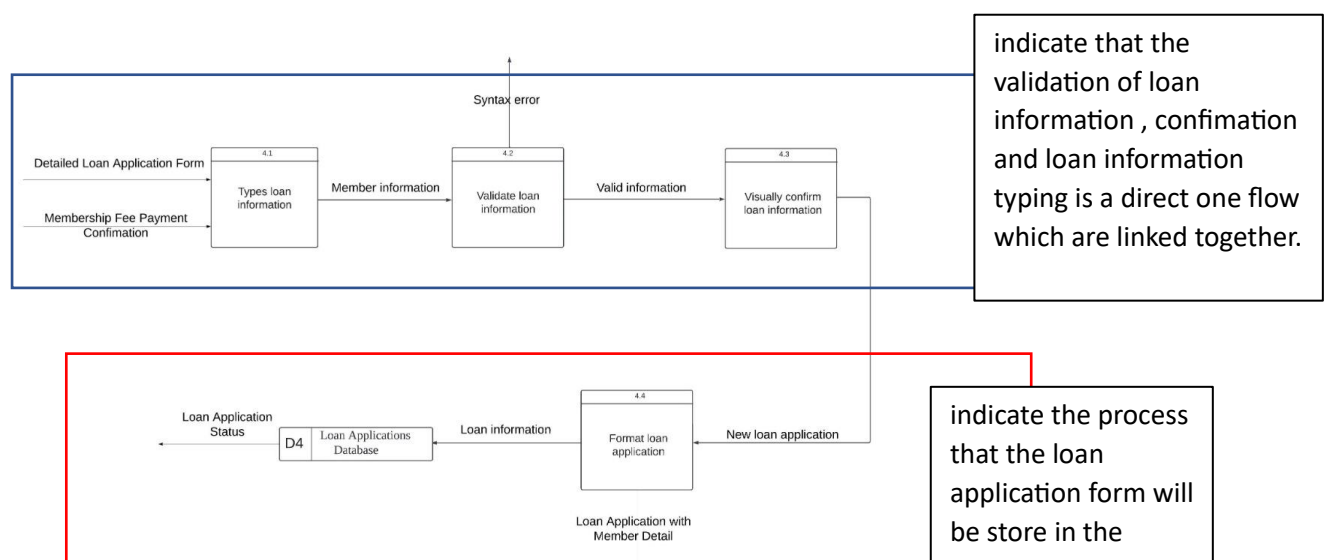
2.3.3 Process 2.0



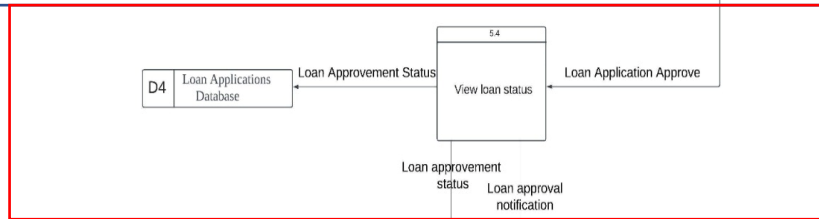
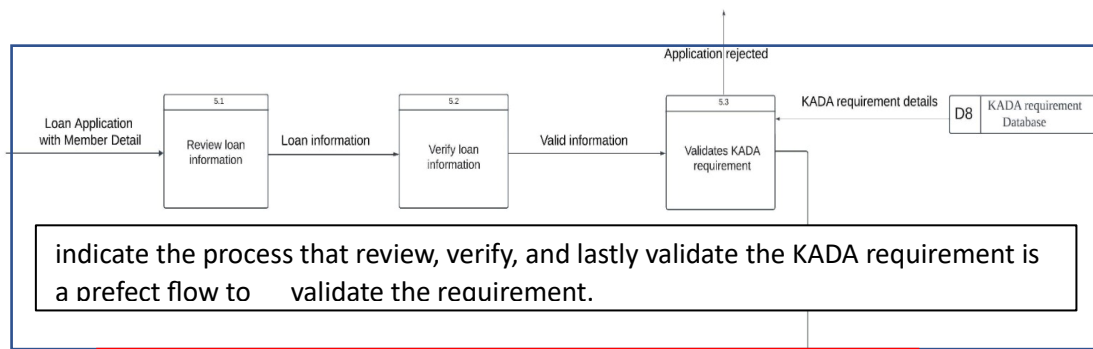
2.3.4 Process 3.0



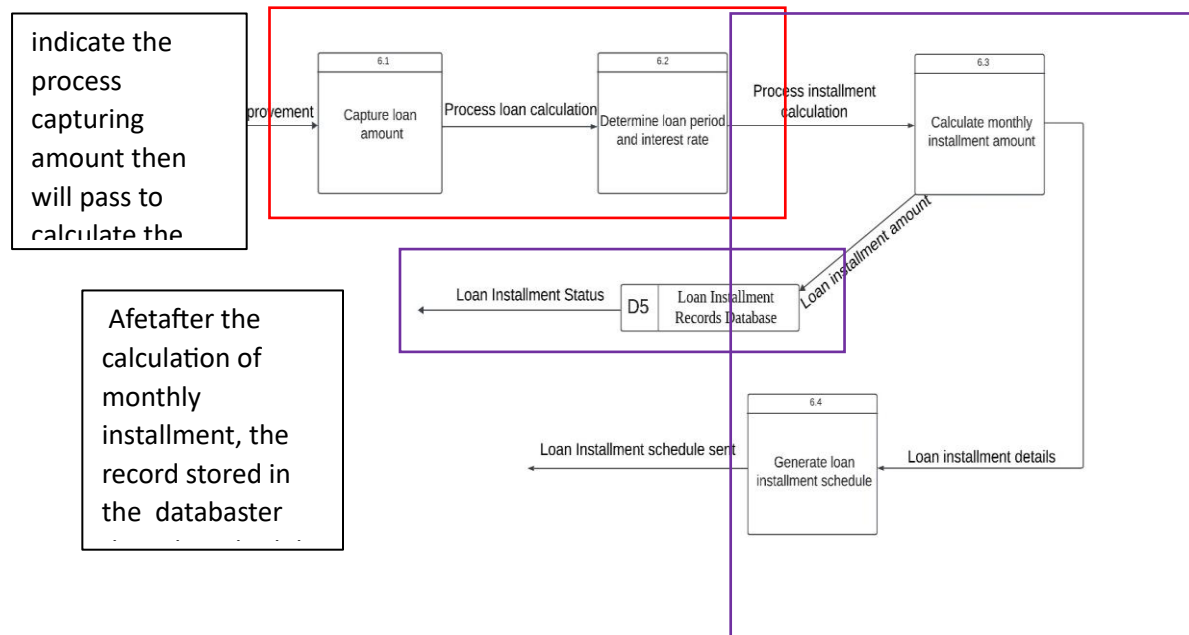
2.3.5 Process 4.0



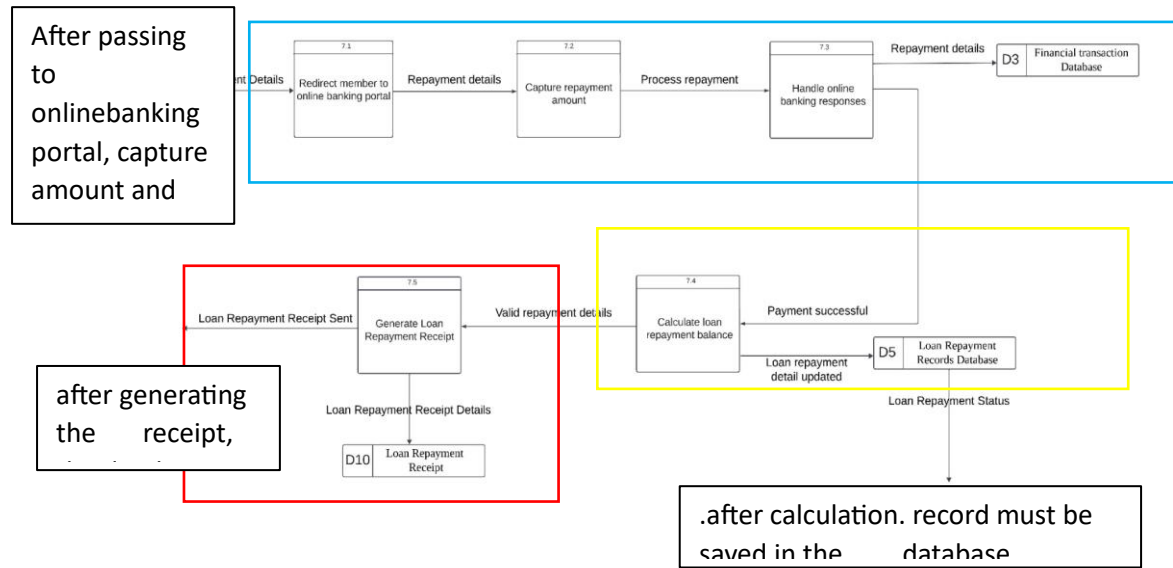
2.3.6 Process 5.0



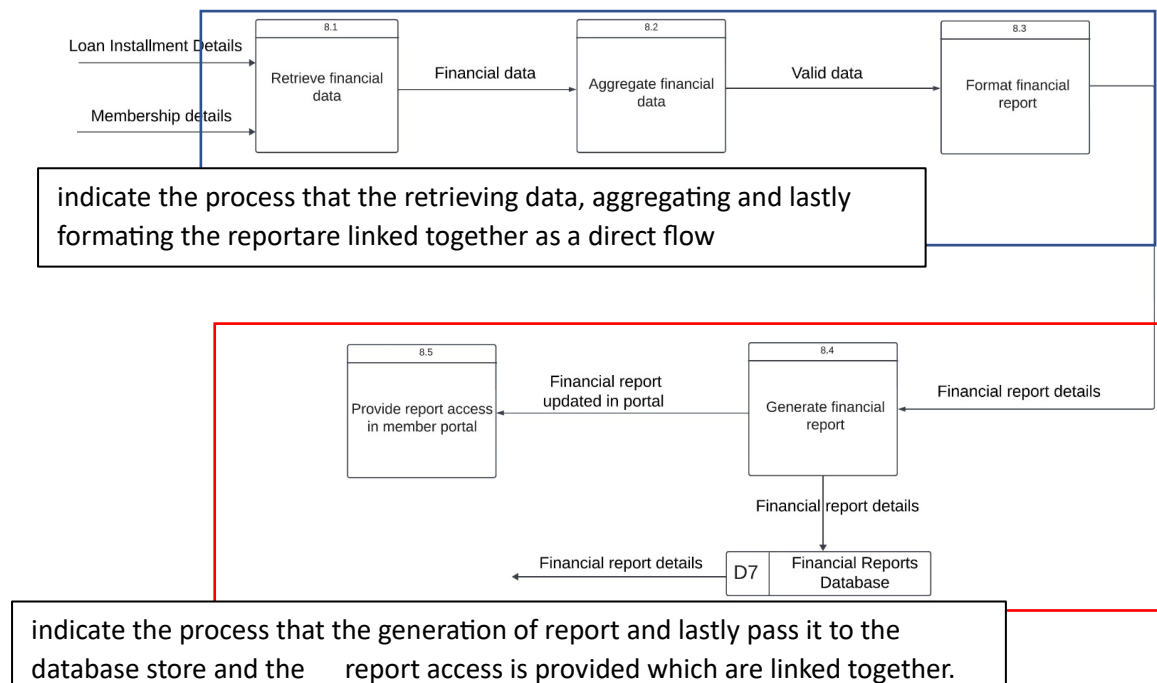
2.3.7 Process 6.0



2.3.8 Process 7.0



2.3.9 Process 8.0



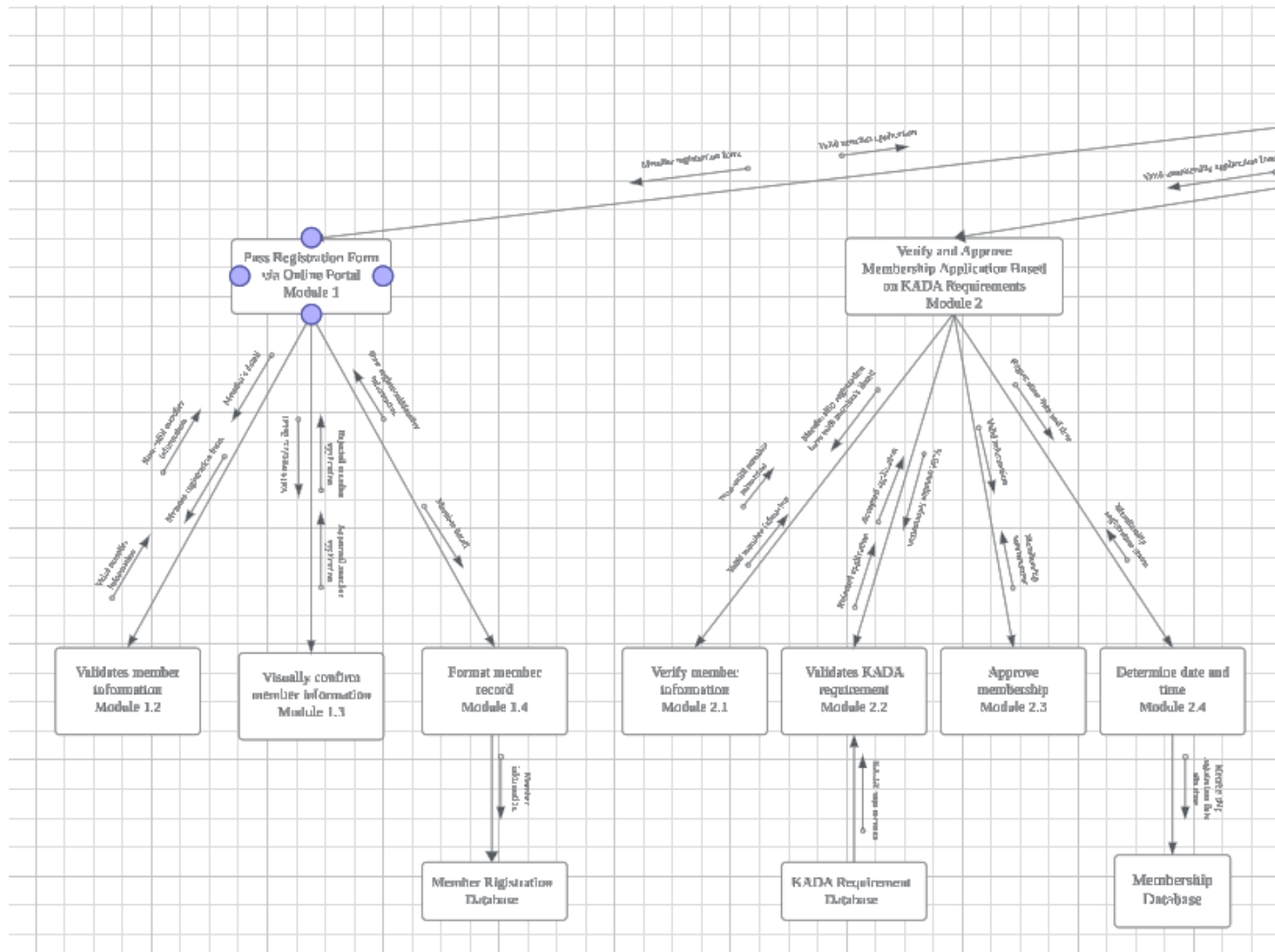
2.4 CRUD Matrix

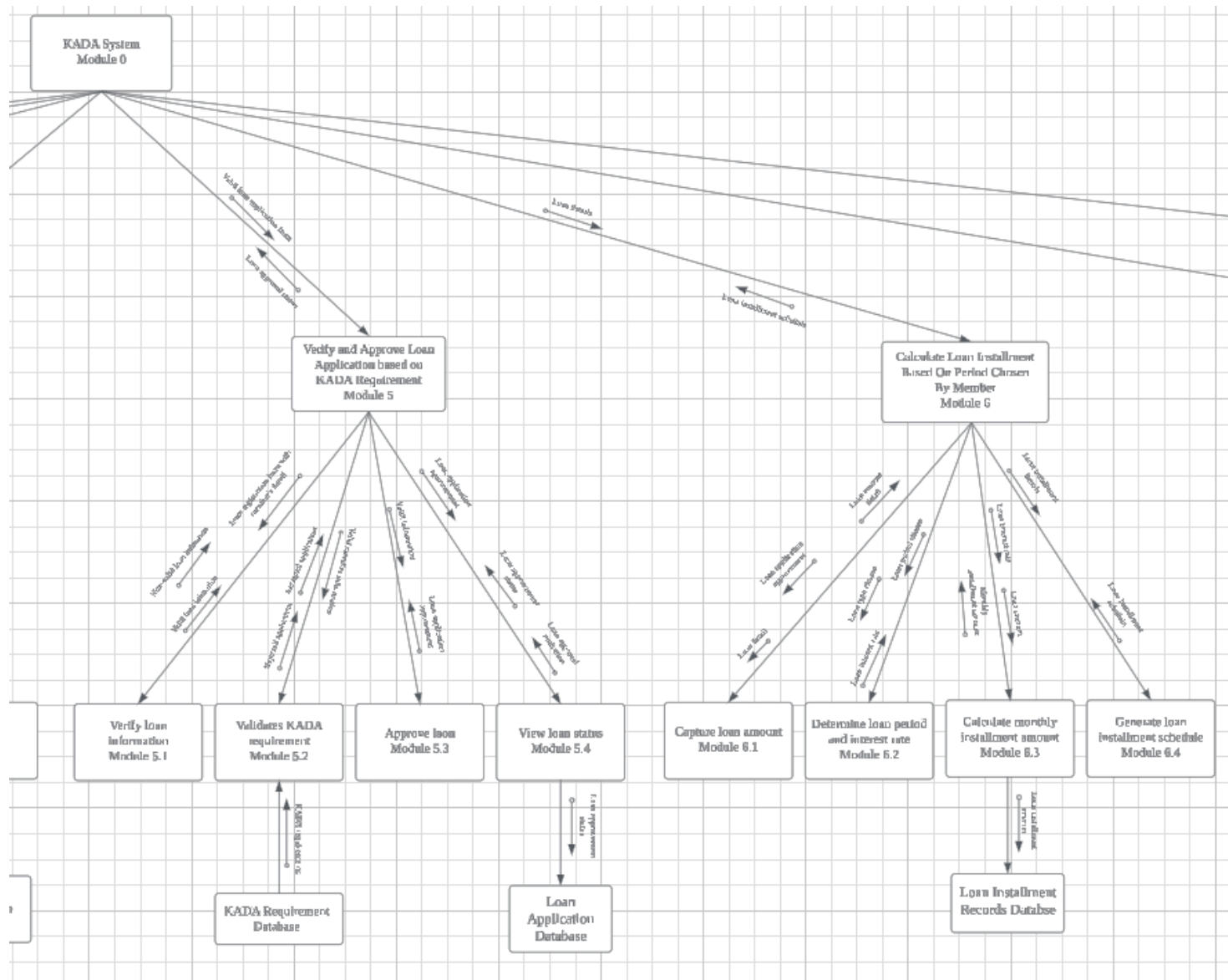
Activity	Member	Admin	Financial Transactions	Loan Application	Loan Instalment Records	Loan Repayment Records	Financial Reports
Pass Registration form	C						
Verify and Approve Membership		C					
Make Membership Fee Payment	R		C				
Submit Loan Application	C			C			
Verify and approve Loan Application		R		RU			
Calculate Loan Instalment				R	C		
Make Loan Repayment	R				R	C	
Generate Financial			R			R	C
Update Application Status	R		U	U	U	U	U

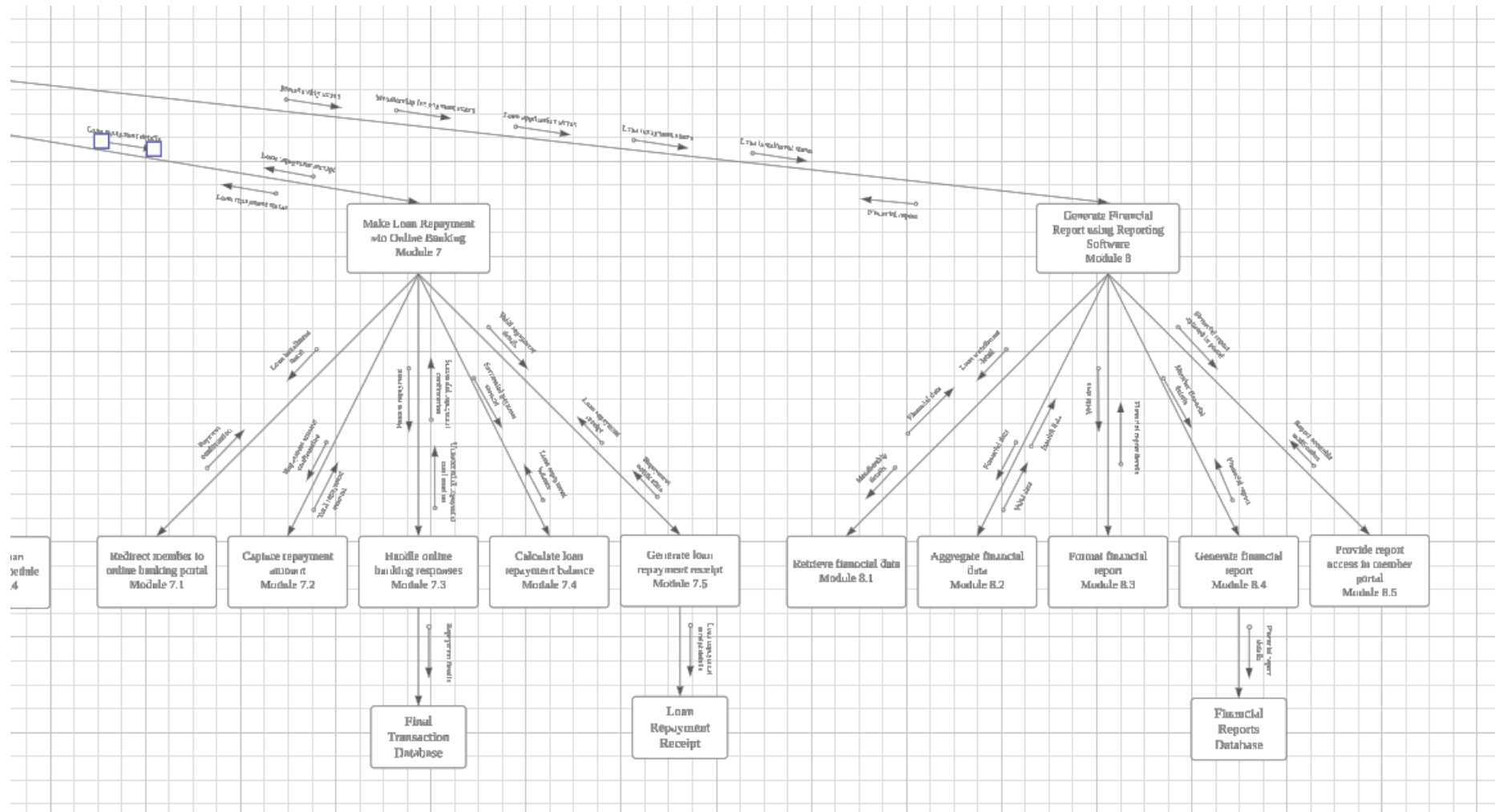
2.5 Event Response Table

Event	Trigger	Activity	Response
Member submit registration form	Detailed member registration form	Validate and store member information	Member information
Membership application approved	Member approvment	Update registration database and notify member	Approval notification
Membership fee payment completed	Membership fee	Update financial transactions database and send receipt	Payment receipt
Loan application submitted	Detailed loan application form	Validate and store loan application information	Loan information
Loan application approved	Loan application with member detail	Update loan application status and notify member	Loan approvment status
Loan installment calculated	Loan application approvment	Store loan installment details	Loan installment schedule
Loan repayment made	Loan repayment	Update loan repayment records and send receipt	Loan repayment receipt
Financial report requested	Loan repayment details	Generate and provide financial report	Financial report

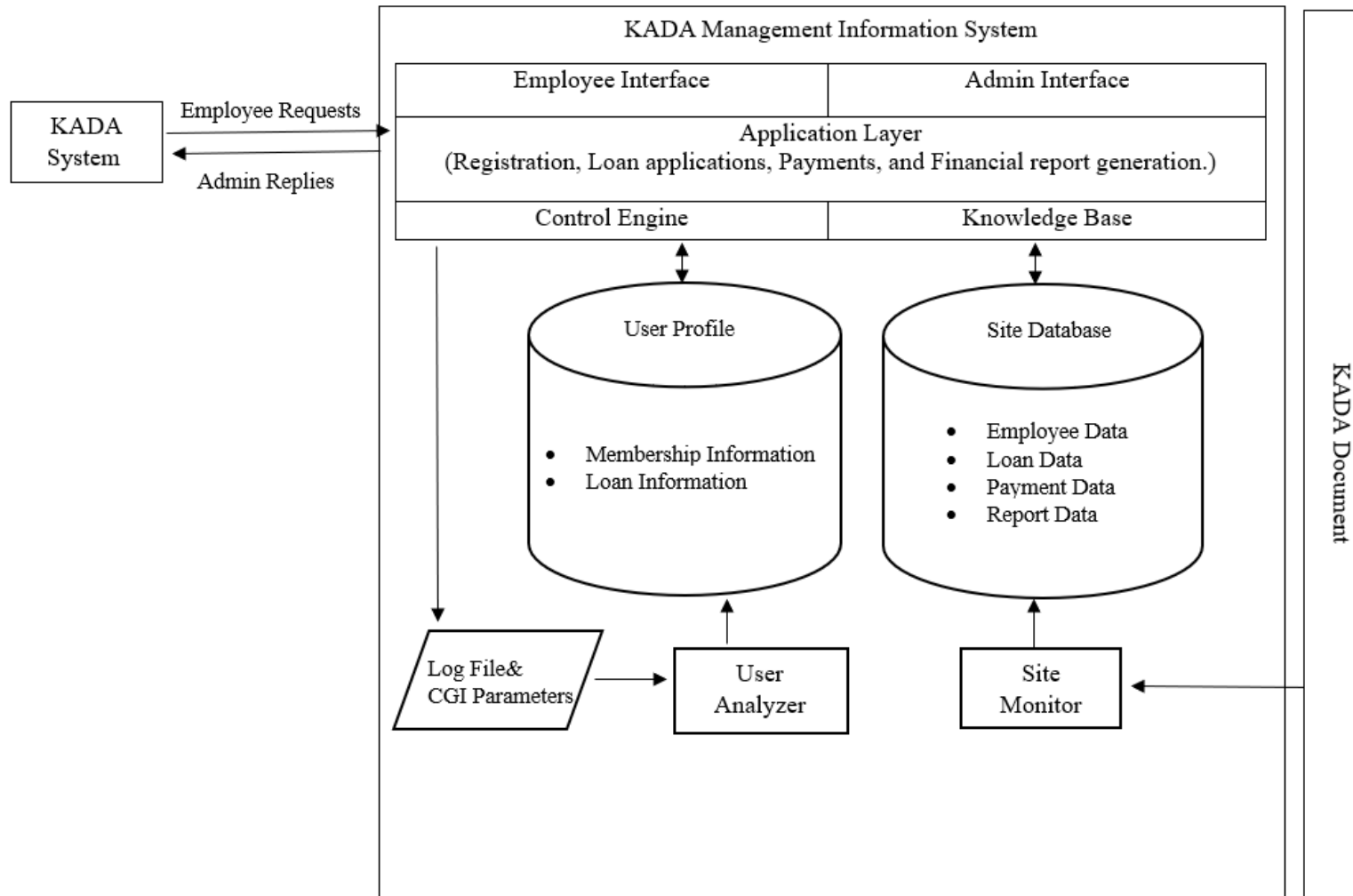
2.6 Structure Chart







2.7 System Architecture



3.0 Wireframe

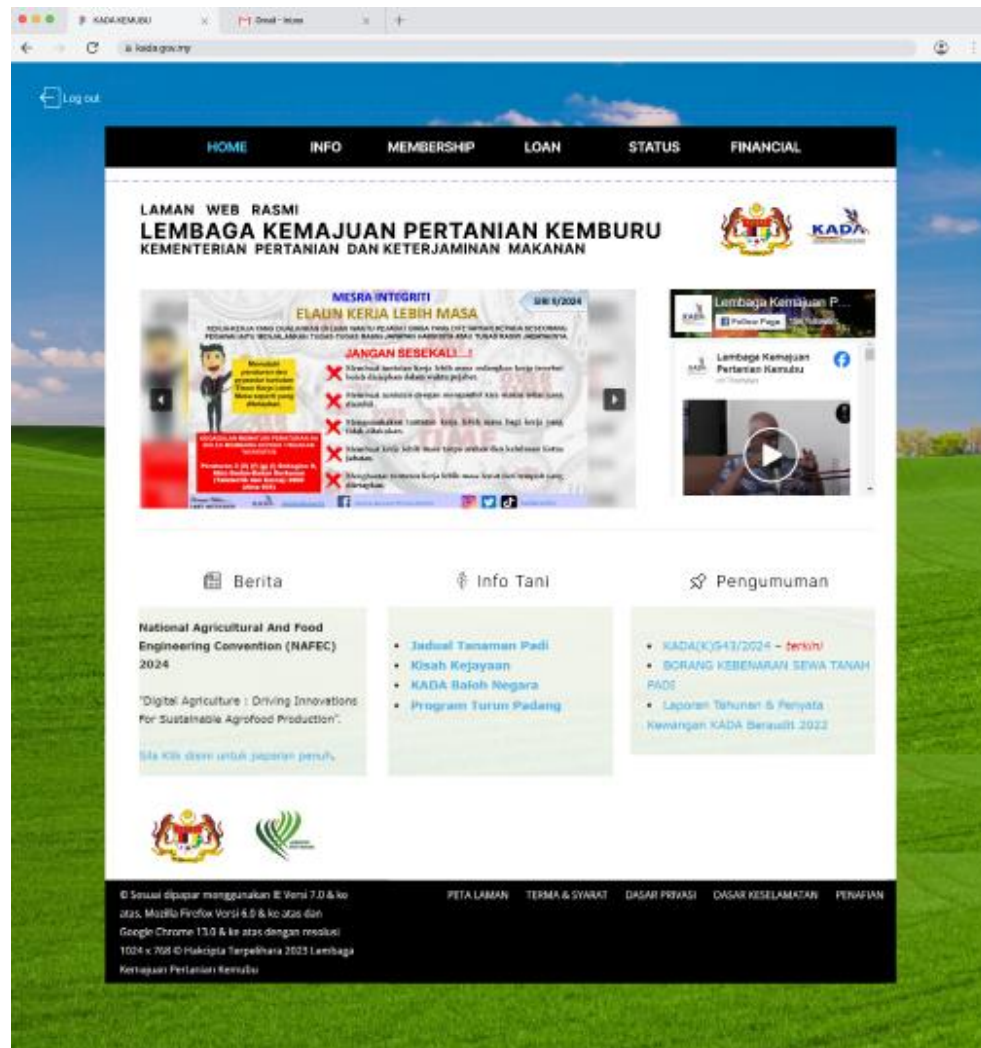
3.1 Input Design

3.1.1 Login Page

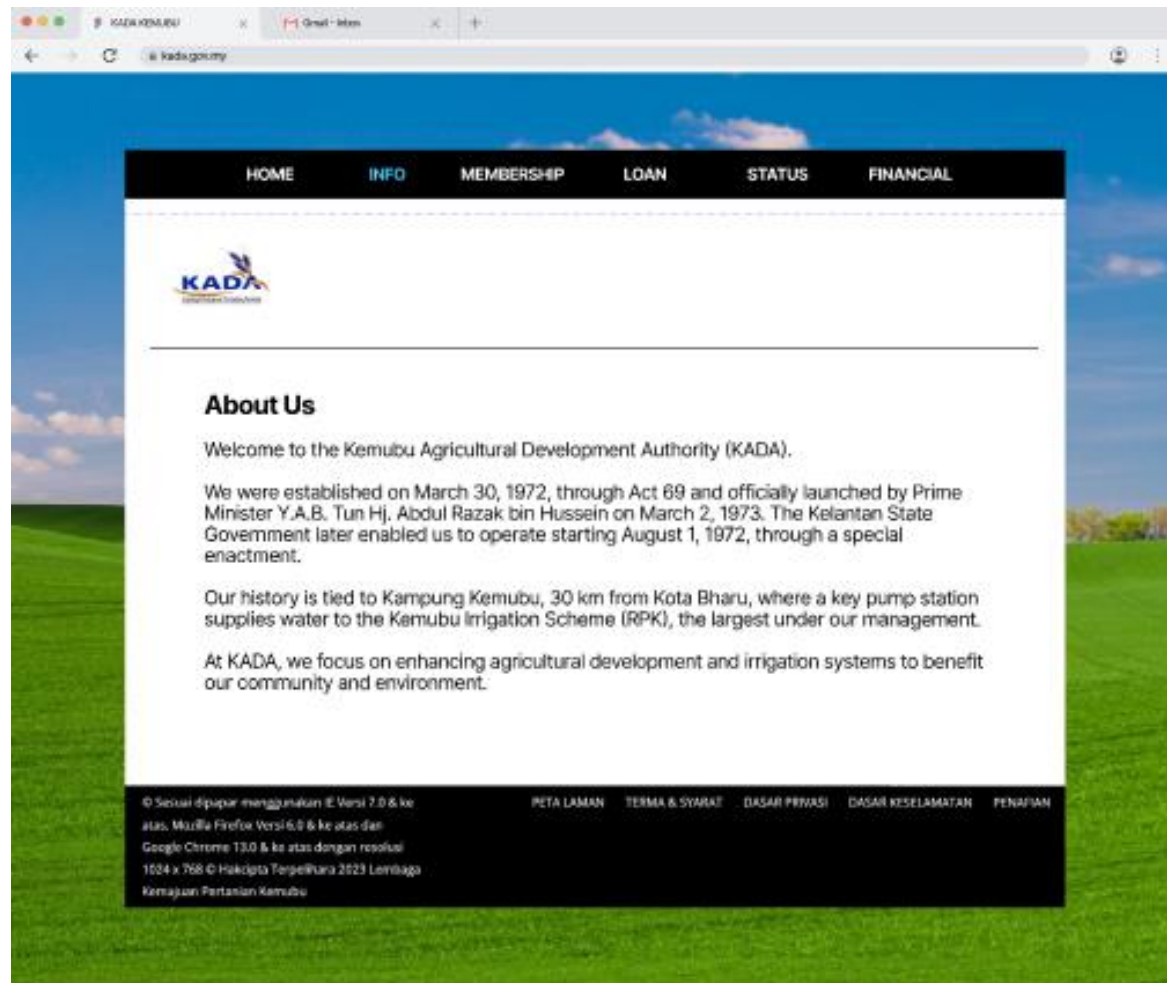
The screenshot shows a web browser window with the URL kada.gov.my. The page features a blue header with the text "MEMBER ADMIN". Below the header is the KADA logo, which includes a stylized bird and the text "KADA" and "KEMAJUAN PERTANIAN KEMBURU". The main content area is white and contains the text "WELCOME TO LEMBAGA KEMAJUAN PERTANIAN KEMBURU". Below this text are two input fields: "Member ID" and "Password", both with placeholder text "Enter your ID/Password". Below the input fields are two buttons: "Login" and "Guest Access". Below the buttons is a link "Forgot Password? CLICK HERE". The footer is black and contains the text "© Sesuai dipapar menggunakan IE Versi 7.0 & ke atas, Mozilla Firefox Versi 6.0 & ke atas dan Google Chrome 13.0 & ke atas dengan resolusi 1024 x 768 © Hakcipta Terpelihara 2023 Lembaga Kemajuan Pertanian Kemubu". The footer also contains a row of links: "PETA LAMAN", "TERMA & SYARAT", "DASAR PRIVASI", "DASAR KESELAMATAN", and "PENAFIAN".

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3.1.2 Homepage



3.1.3 Information Page



3.1.4 Member Registration Interface -Personal Information

KADA KEMASU Gmail - Inbox kada.gov.my

HOME INFO **MEMBERSHIP** LOAN STATUS FINANCIAL

PERMOHONAN MENJADI ANGGOTA

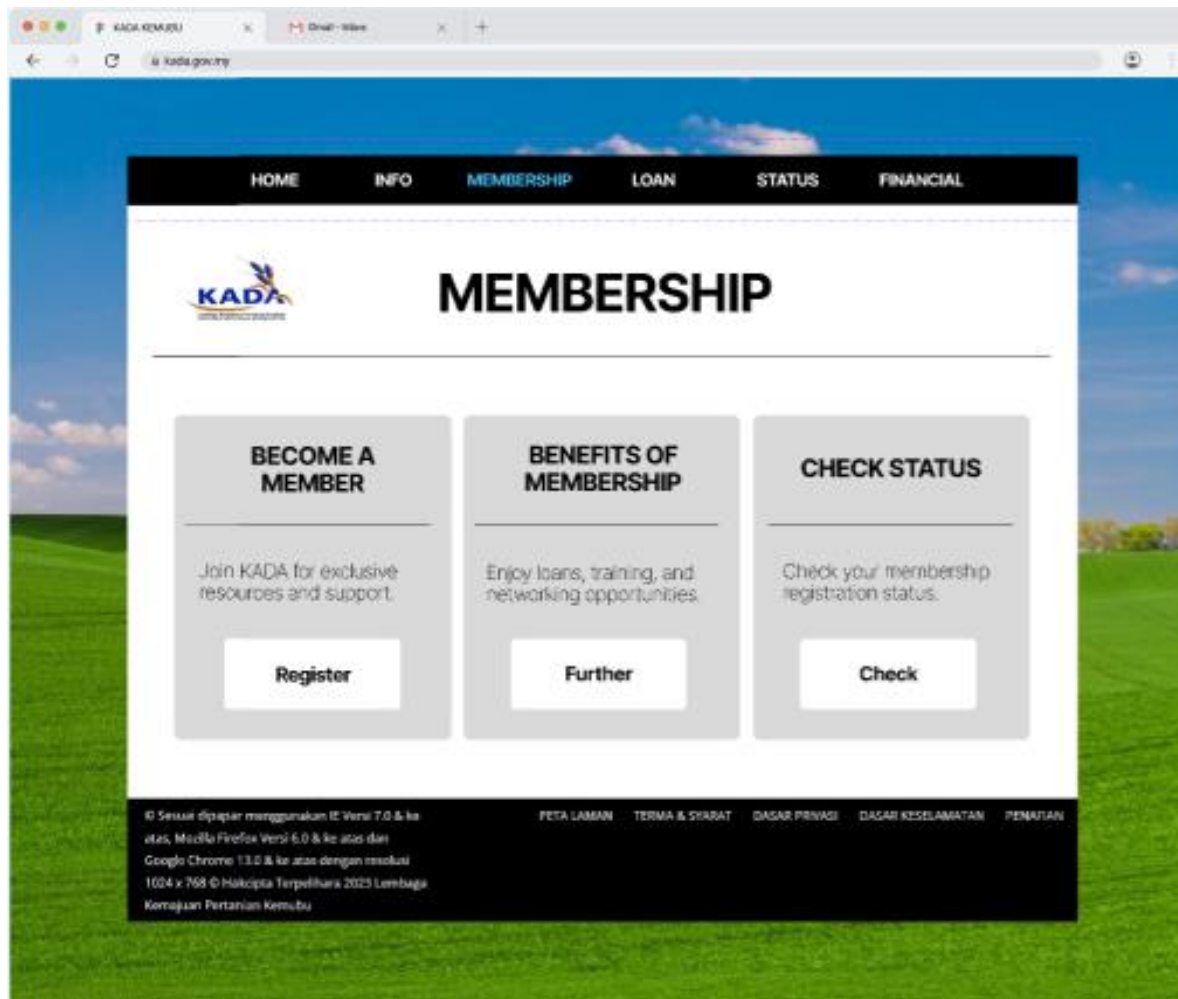
MAKLUMAT PERIBADI	ALAMAT	MAKLUMAT PEKERJAAN	MAKLUMAT HUBUNGAN
NAMA			
NO. KP			
TARAF PERKAHWINAN	Pilih	BANGSA	Pilih

BACK NEXT

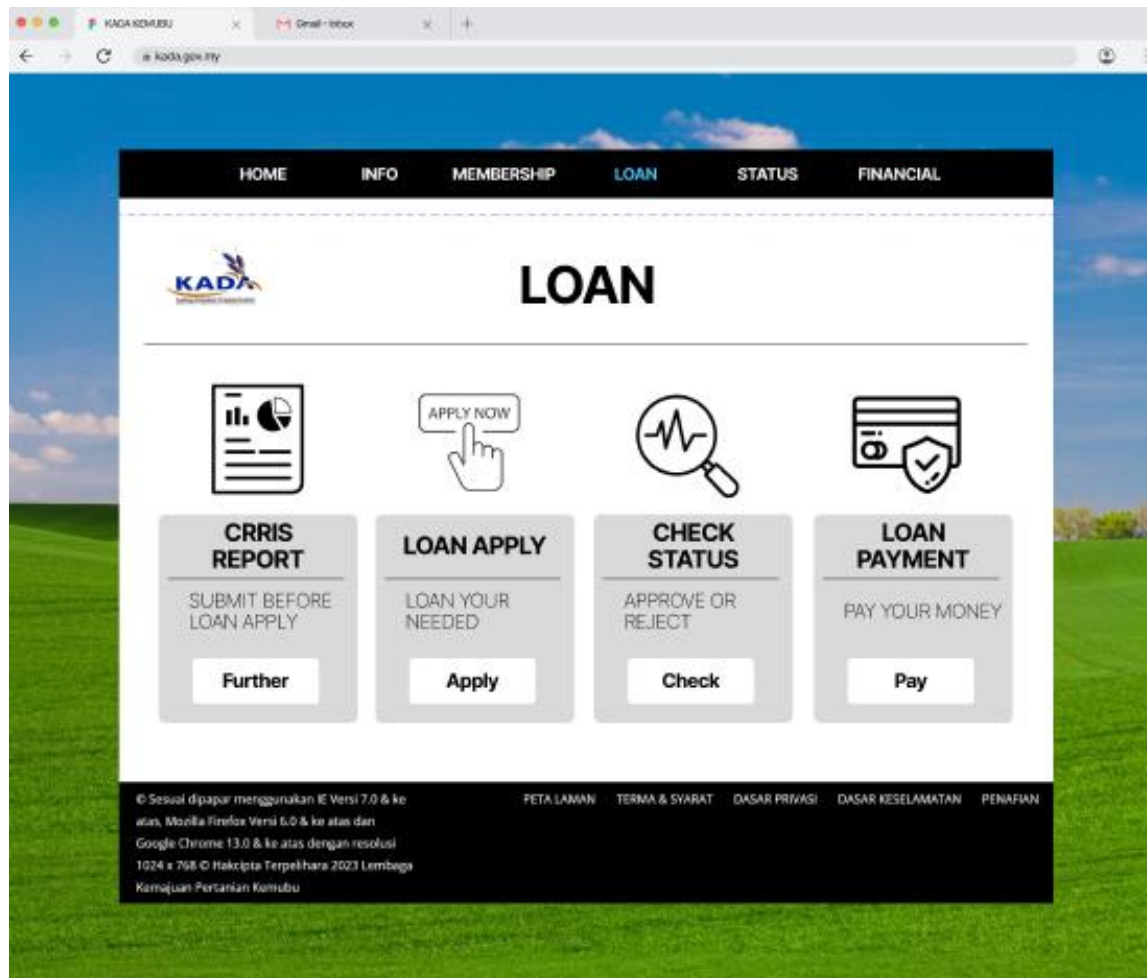
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PETA LAMAN TERMA & SYARAT DASAR PRIVASI DASAR KESELAMATAN PENAFIAN

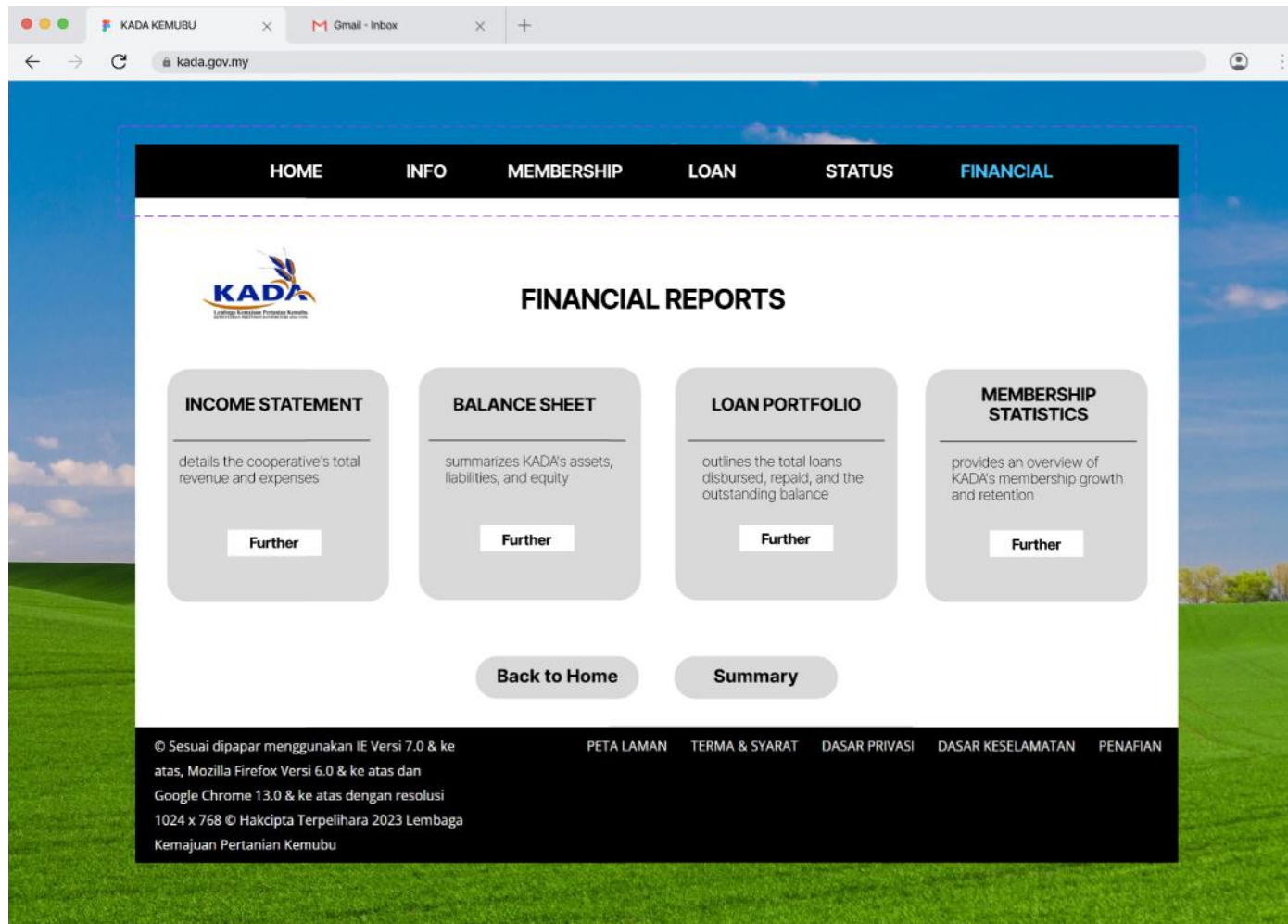
3.1.5 Membership Home Page



3.1.6 Loan Application Home Page



3.1.7 Financial Information Home Page



3.2 Output Design

3.2.1 Status Dashboard

The screenshot shows a web browser window with the URL kada.gov.my. The page features a navigation bar with links: HOME, INFO, MEMBERSHIP, LOAN, STATUS, and FINANCIAL. The main content area is titled "Status Dashboard" and includes the KADA logo. Below the title, the user's name "Ali Bin Abu" and Member ID "66443215" are displayed. A table lists the user's application history, showing details for Membership and Loan applications, including their status (Approved, Rejected, Pending) and a link to view details. A "BACK TO HOME" button is located below the table. The footer contains copyright information and links to PETA LAMAN, TERMA & SYARAT, DASAR PRIVASI, DASAR KESELAMATAN, and PENAFIAN.

NAME : Ali Bin Abu
Member ID : 66443215

Application Type	Application ID	Application Date	Status	Action
Membership	315645	1 / 4 / 2022	Approved	[View]
Loan	3254123	14 / 6 / 2022	Approved	[View]
Loan	12514124	9 / 9 / 2022	Rejected	[View]
Loan	12425231	14 / 9 / 2022	Approved	[View]
Loan	42415412	24 / 4 / 2023	Approved	[View]
Loan	34562121	12 / 1 / 2024	Approved	[View]
Loan	16242423	15 / 6 / 2024	Pending	[View]

[BACK TO HOME](#)

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[PETA LAMAN](#) [TERMA & SYARAT](#) [DASAR PRIVASI](#) [DASAR KESELAMATAN](#) [PENAFIAN](#)

3.2.2 Payment Interface

The screenshot shows a web browser window with the URL kada.gov.my. The page has a blue header with navigation links: HOME, INFO, MEMBERSHIP, LOAN, STATUS, and FINANCIAL. The main content area is titled "Registration and Investment Payment" and features the KADA logo. It displays the following information:

Total : RM 355
Name : Ali Bin Abu
Member ID : 66443215

Choose preferred payment method:

There are two payment options, each with a radio button:

- ☒ VISA CREDIT CARD
- ☐ FPX BANKING

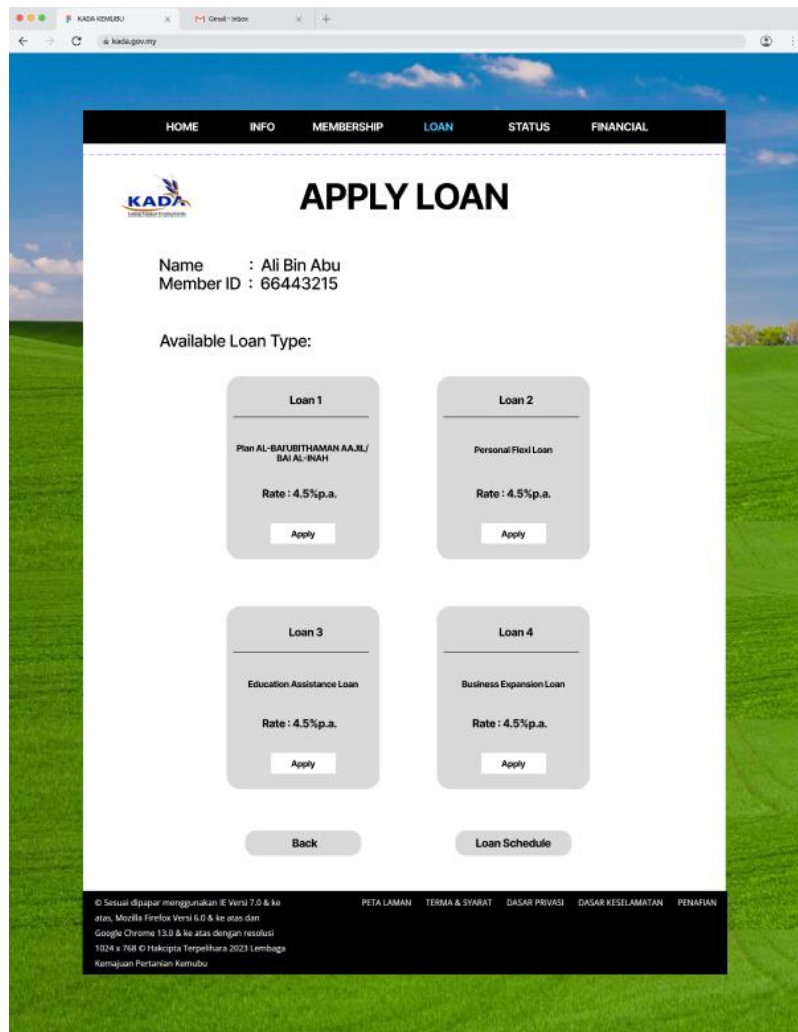
At the bottom of the main content area are two buttons: "Back" and "Next".

The footer contains the following text:

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Navigation links in the footer: PETA LAMAN, TERMA & SYARAT, DASAR PRIVASI, DASAR KESELAMATAN, PENAFIAN

3.2.3 Available Loan Type Interface



3.2.4 Loan Payment Schedule

KADA KEMBU
Gmail - Inbox
kda.gov.my

HOME
INFO
MEMBERSHIP
LOAN
STATUS
FINANCIAL


KADA
KEMAJUAN AGRIKULTUR DAN AGROUMESTIA

Loan Repayment Schedule

Rate of interest		4.2 %					
Period (YEAR)	1	2	3	4	5	6	
Period (MONTHLY)	12	24	36	48	60	72	
Total Financing			Monthly installment				
1000.00	86.83	45.17	31.28	24.33	20.17	17.39	
2000.00	173.67	90.33	62.56	48.67	40.33	34.78	
3000.00	260.50	135.50	93.83	73.00	60.50	52.17	
4000.00	347.33	180.67	125.11	97.33	80.67	69.56	
5000.00	434.17	225.83	156.39	121.67	100.83	86.94	
6000.00	521.00	271.00	187.67	146.00	121.00	104.33	
7000.00	607.83	361.17	218.94	170.33	141.17	121.72	
8000.00	694.67	361.33	250.22	194.67	161.33	139.11	
9000.00	781.50	406.50	281.50	219.00	181.50	156.50	
10000.00	866.33	451.67	312.78	243.33	201.67	173.89	
11000.00	955.17	496.83	344.06	267.67	221.83	191.28	
12000.00	1042.00	542.00	375.33	292.00	242.00	208.67	
13000.00	1128.83	587.17	406.61	316.33	262.17	226.06	
14000.00	1215.67	632.33	437.89	340.67	282.33	243.44	
15000.00	1302.50	677.50	469.17	365.00	302.50	260.83	
16000.00	1389.33	722.67	500.44	389.33	322.67	278.82	
17000.00	1476.17	767.83	531.72	413.67	342.83	195.61	
18000.00	1563.00	813.00	563.00	438.00	363.00	313.00	
19000.00	1649.83	858.17	594.28	462.33	383.17	275.06	
20000.00	1736.67	903.33	625.56	486.67	403.33	347.78	

Back

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PETA LAMAN TERMA & SYARAT DASAR PRIVASI DASAR KESELAMATAN PENAFIAN

3.2.5 Financial Report Summary

